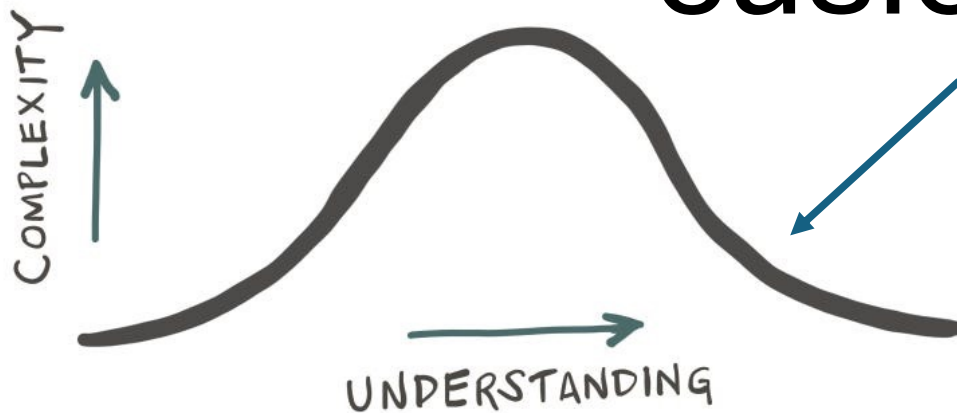


“as a system evolves, its complexity increases unless work  
is done to reduce it” — Lehman, 1980 [1]  
if anyone can do it for \$1 ... can you do it for a cent?



# easier AI?



[timmm@ieee.org](mailto:timmm@ieee.org)

ssbse'26: july 6, 2026

code: **pip install ezr**  
data: **<http://tiny.cc/moot>**





Ssbse'16

[2]

## An (Accidental) Exploration of Alternatives to Evolutionary Algorithms for SBSE

Vivek Nair<sup>(✉)</sup>, Tim Menzies, and Jianfeng Chen

North Carolina State University, Raleigh, USA  
vivekax1@gmail.com

**Abstract.** SBSE researchers often use an evolutionary algorithm to solve various software engineering problems. This paper explores an alternate approach of sampling. This approach is called SWAY (Sampling WAY) and finds the (near) optimal solutions to the problem by (i) creating a larger initial population and (ii) intelligently *sampling* the solution space to find the best subspace. Unlike evolutionary algorithms, SWAY does not use mutation or cross-over or multi-generational reasoning to find interesting subspaces but relies on the underlying dimensions of the solution space. Experiments with Software Engineering (SE) models shows that SWAY's performance improvement is competitive with standard MOEAs while, terminating over an order of magnitude faster.

**Keywords:** Search-based SE · Sampling · Evolutionary algorithms



- 2019:  
**Amritanshu Agrawal** (Snap);  
**Vivek Nair** (Meta)  
**Jianfeng Chen** (Meta)
- 2025:  
**Andre Lustosa** (Redhat)
- 2027:  
**Kishan Ganguly** BINGO (FSE'26);  
**Amirali Rayegon** Explanation (JSS'26)
- 2028:  
**Srinath Srinivasan**, Neurosymbolic (arxiv)

**why easier AI?**

how to do easier AI?

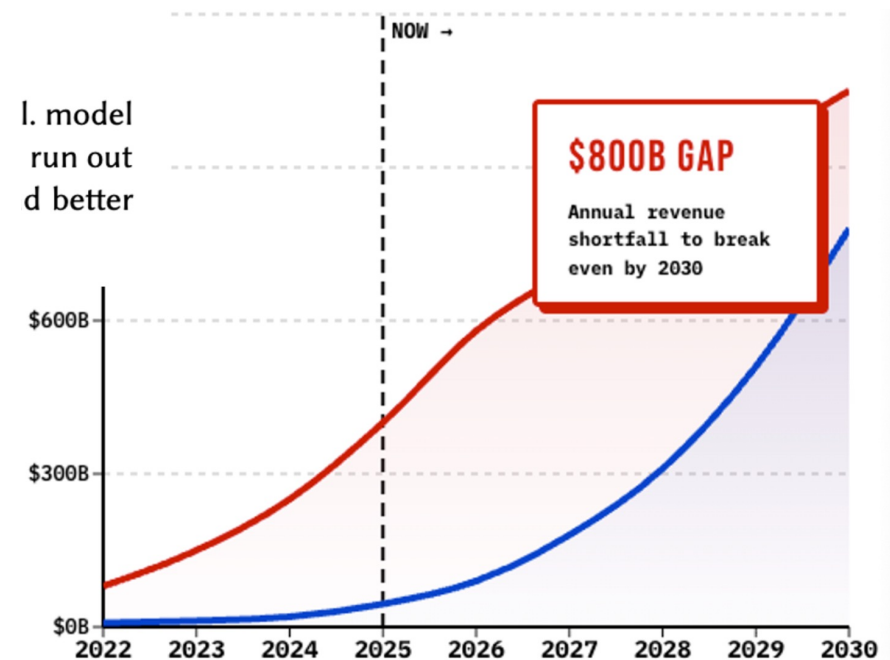
why not more of easier AI?



# Q: why easier AI?

## A: economics [3] [4]

- Niklaus Wirth:
  - “Software gets **slower** faster than hardware gets faster”
- Even if \$/token falls, Jevons paradox: total token burn grows (\*10, \*100: 2023→2026)
- Bain 2025 forecast:
  - \$800B/yr revenue gap by 2030
  - 2026 capex ~\$725B; 2027 projected >\$1T — gap widening



Q: why easier AI?

A: easier to trust; easier to watch and understand



## My Tesla Was Driving Itself Perfectly —Until It Crashed

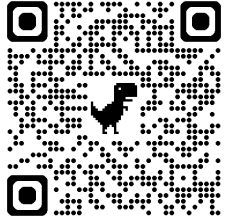
The danger of almost-perfect tech

By Raffi Krikorian



THE **ATLANTIC**  
MARCH 17, 2026



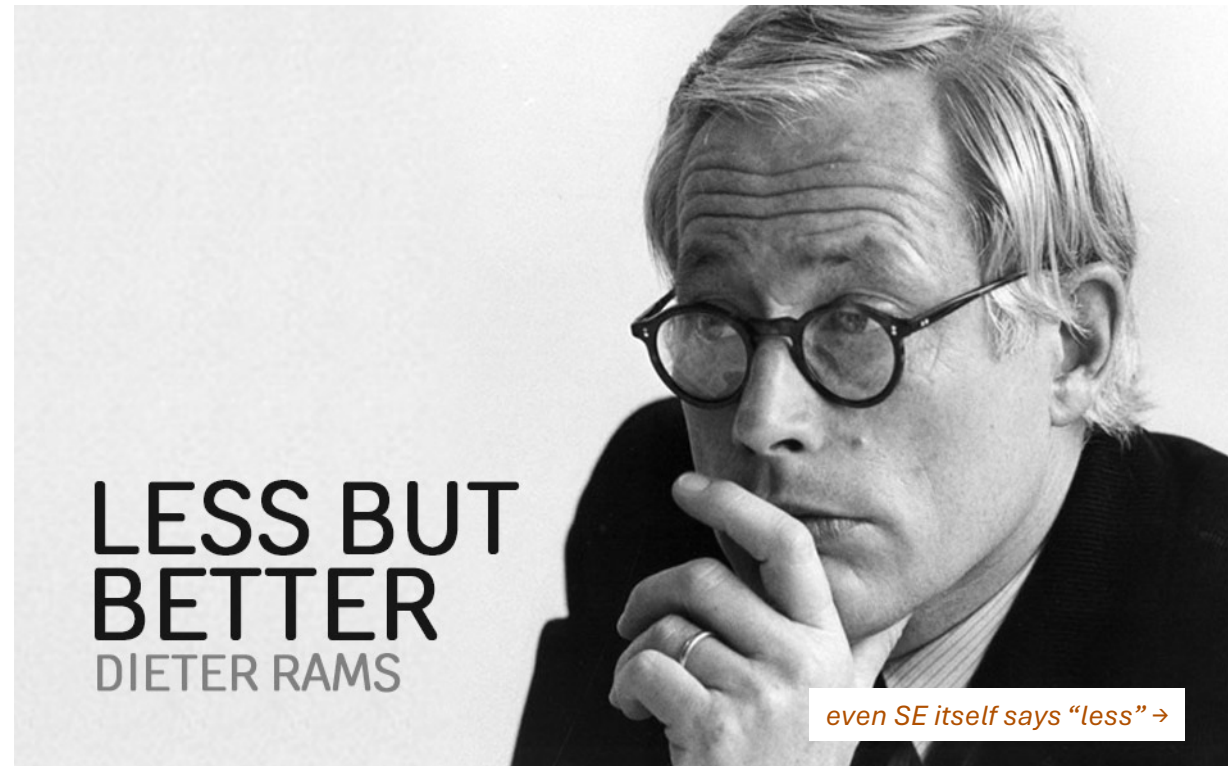


Q: why easier AI?

A: cause some domains demand it [5]

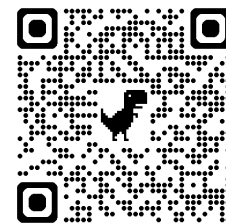
- Green AI
- global **south** (countries without e.g. American-scale infrastructure)
- resource-constraint teaching
  - community colleges,
  - K-12 CS,
  - MSIs,
  - thise using **personal** hardware
- privacy-constrained industrial dev
  - **defense**,
  - regulated healthcare,
  - financial services,
  - law firms)
- air-gapped settings;
- **edge** or on-device inference

6

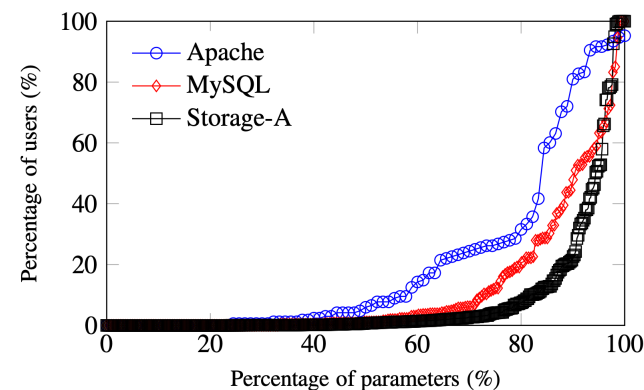
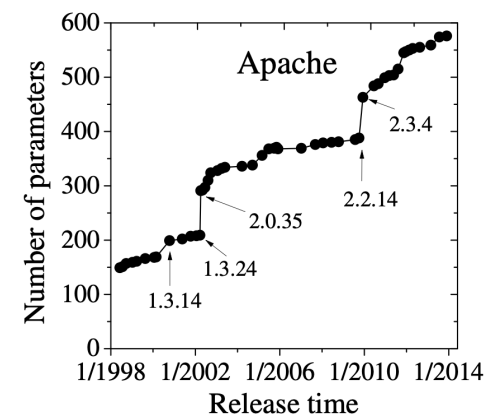


# Q: why easier AI?

## A: software engineering



- **security:**
  - modern software = assembly of pre-made parts (> 80%)
  - more reuse == more exposure
  - e.g. left-pad; MOVEit (\$12B), Falcon (\$5B), ByBit (1.4B) \$24B (2025);
- **survivability:** e.g. SQLite does not use Berkeley DB.
  - Now used on Mars
- **maintainability + reliability:**
  - more loc = more to maintain, more to go wrong
- **waste:** Most users cannot handle most options [6]



and the data for “big” is running out →

Q: why easier AI?

A: cause “big AI” needs too much data [7,10]

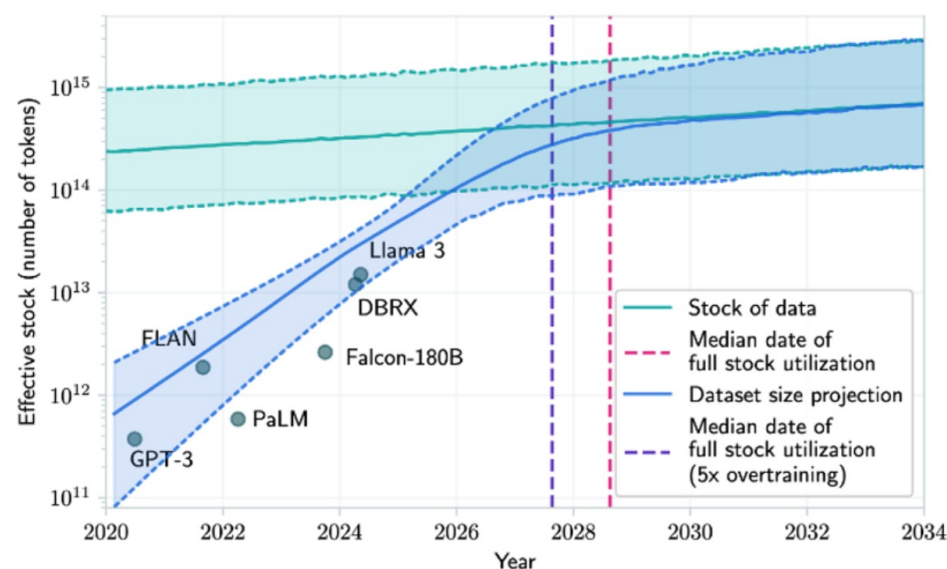
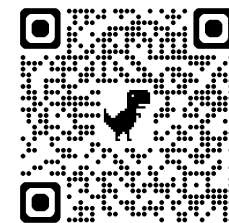
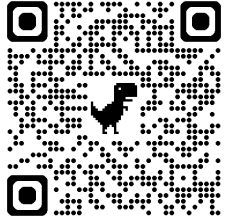


Fig. 4. Median results from the Villalobos et al. model (shown in green) estimate that by 2028, we will run out of new textual data needed to train bigger and better LLMs [75].

*besides, more was never always better →*





Q: why easier AI?

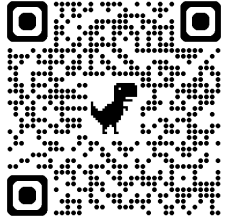
A: more AI not always better [8][9][10]

- text mining predicting effort within **agile** projects,
  - (SVM and TFIDF) better predictions than deep learners
  - Runs 100 times faster
- tree-based models (Random Forest, XGBoost)
  - consistently **outperform** deep neural networks on tabular datasets,
  - especially those with more than 10,000 rows
- Etc [11-14]

why easier AI?

**how to do easier AI?**

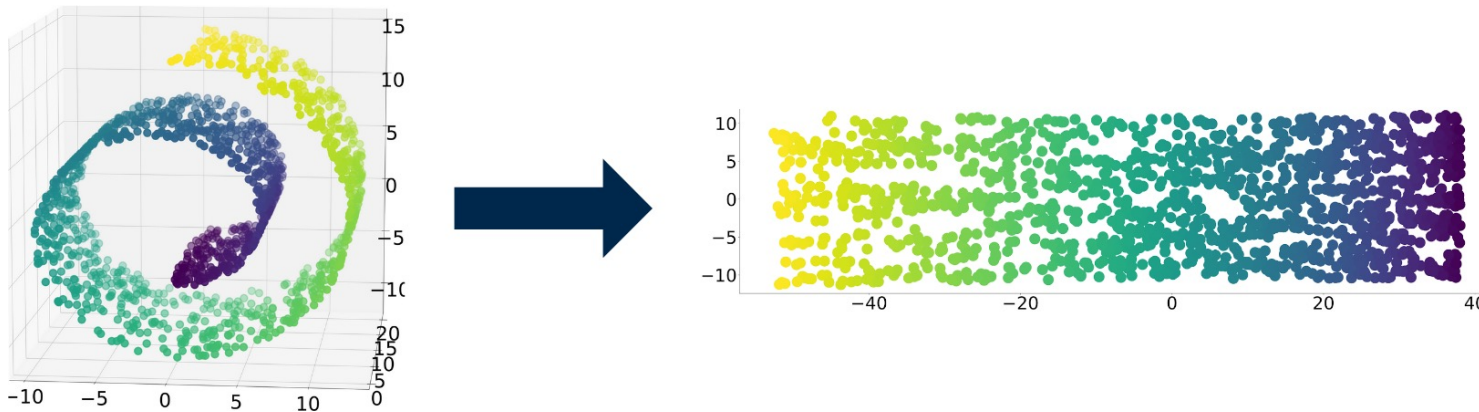
why not more of easier AI?

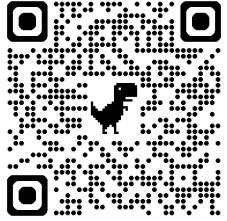


# Q: how to do easier AI?

## A: so many ways [15-28]

- 1300: William of Ockham: “entities must not be multiplied beyond necessity”
- 1902, **PCA**: reduce data to a few principal components
- 1974, **Prototypes**: speed up k-NN by reducing rows to a few exemplars
- 2009, **Active learning**: only use most informative rows
- 2020s, Distillation: compress large LLM models with little performance loss
- etc

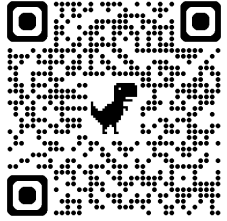




Q: how to do easier AI (with active learning)

A: use current model to decide what's next [29]

- E.g. **SMAC3**
  - ensemble of decision trees
  - Seek unlabeled examples where
    - ensemble most disagrees (a.k.a. explore)
    - **ensemble** most agree (a.k.a. exploit)
    - adapt =  
Slowly shift explore to exploit
- Cost of SMAC3:
  - when new data arrives
  - **rebuilding** the ensemble
- EZR: **pip install ezr**
- “ensemble” = 2 **class** Bayes classifier
  - $|Best|, |Rest| = \sqrt{N}, N - \sqrt{N}$ ,
  - **next** =
    - closest to *Best*
    - furthest from *Rest*
  - update:
    - add next to *Best*,
    - resort *Best*,
    - **pop** worst *Best* to *Rest*
  - Very fast
    - never sort *Rest*
    - Updates to *Best* and *Rest* are fast.



Q: how to test EZR?

A: read lots of papers, collect their data [30]

- Recent SBSE **papers**: ICSE, FSE, TSE, IST, EMSE, TOSEM, ASE
- 127 data sets
- 3-1044 independent **x attributes**;
- 1 to 8 dependent **y attributes** to be minimized or maximized;
- 93 to 100,000 rows.
- <http://tiny.cc/moot>

| #                                                        | Category      | Focus    | File Name(s)     | Here, optimizers struggle with issues like...       | #y  | #x     | Rows     | Refs |
|----------------------------------------------------------|---------------|----------|------------------|-----------------------------------------------------|-----|--------|----------|------|
| <b>Software Systems &amp; Configuration &amp; Tuning</b> |               |          |                  |                                                     |     |        |          |      |
| 24                                                       | Soft. Config  | PLE      | SS-[A-X]         | Minimize footprint/memory vs. maximizing throughput | 2-3 | 3-88   | 197-86k  | [1]  |
| 12                                                       | PromiseTune   | Perf.    | 7z, LLVM, BDBC   | Minimize execution time and energy consumption      | 1   | 6-35   | 864-166k | [2]  |
| 1                                                        | Cloud compute | Tuning   | Apache_AllMeas   | Balance server response vs. CPU/RAM load patterns   | 1   | 9      | 192      | [3]  |
| 1                                                        | Cloud compute | Tuning   | SQL_AllMeas      | Minimize latency/IO vs. maximizing throughput       | 1   | 39     | 4,654    | [4]  |
| 1                                                        | Cloud compute | Tuning   | X264_AllMeas     | Optimize encoding parameters for PSNR/SSIM quality  | 1   | 16     | 1,153    | [1]  |
| 2                                                        | Testing       | Testing  | test120, 600     | Maximize coverage while minimizing execution time   | 1   | 9      | 5,161    | -    |
| <b>Project Management &amp; Process Modeling</b>         |               |          |                  |                                                     |     |        |          |      |
| 35                                                       | Proj. Health  | Health   | Health-Closed    | Optimize PR rates and minimize developer churn      | 2-3 | 5      | 10,001   | [5]  |
| 3                                                        | Scrum         | Agile    | Scrum[1k-100k]   | Maximize velocity within sprint constraints         | 3   | 124    | 1k-100k  | [6]  |
| 8                                                        | Feature Mod.  | Config   | FFM, FM-*        | Optimize Clause/Constraint ratio in large spaces    | 3   | 128-1k | 10,001   | [1]  |
| 1                                                        | nasa93dem     | Cost     | nasa93dem        | Minimize effort (person-months) vs. quality         | 3   | 26     | 93       | [5]  |
| 1                                                        | COC1000       | Risk     | coc1000          | Minimize risks/schedule slip vs. analyst expertise  | 5   | 20     | 1,001    | [7]  |
| 4                                                        | POM3          | Process  | pom3[a-d]        | Balance project idle rates vs. completion costs     | 3   | 9      | 501-20k  | [6]  |
| 4                                                        | XOMO          | Defects  | xomo_[flt,grd]   | Minimize defect density vs. total project effort    | 4   | 27     | 10,001   | [7]  |
| <b>Interdisciplinary &amp; Behavioral Benchmarks</b>     |               |          |                  |                                                     |     |        |          |      |
| 4                                                        | Behavioral    | Behavior | all_players, etc | Maximize user retention and predict cohort churn    | 1-3 | 26-55  | 82-17k   | [8]  |
| 4                                                        | Financial     | Finance  | BankChurners     | Minimize credit risk vs. optimal pricing models     | 2-5 | 19-77  | 1k-20k   | [9]  |
| 3                                                        | Health        | Health   | COVID19, Life    | Maximize accuracy/life vs. readmission likelihood   | 1-3 | 20-64  | 2k-25k   | [10] |
| 2                                                        | RL            | Control  | A2C_[Acr,Crt]    | Maximize reward signals vs. steps to convergence    | 3-4 | 9-11   | 224-318  | -    |
| 5                                                        | Sales         | Market   | Marketing, socks | Maximize ROI vs. minimizing forecasting error       | 1-8 | 20-31  | 247-2k   | [11] |
| 3                                                        | Misc.         | General  | auto93, Car      | Optimize multivariate trade-offs (e.g. MPG vs. HP)  | 2-5 | 5-38   | 205-1k   | [1]  |

configuration, performance tuning, product lines, project health, defect prediction, testing, cost estimation, cross-domain generalization, and text mining

... and a score →



Q: how to score EZR?

A: win = 1 - normalize(regret) [31]

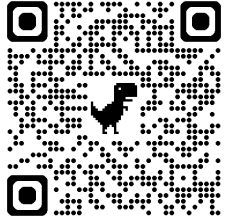
- Distance to **heaven** (aggregate n goals to one)

$$d2h(y) = \sqrt{\sum_{i=1}^k y_i^2 / \sqrt{k}},$$

- Regret = (**d2h** - lo) / (mu - lo)
- Win = int(100 \* (1 - regret))
  - Win=0 → optimizer just finds mu
    - Failure**
  - Win=100 → found reference optimum
    - Success
- Build regression tree on **labeled** eggs
  - Splitting rows to minimize  $\sigma$  of win
  - Reports what takes you towards good or bad

| win   | n  | Lbs-   | Acc+ | Mpg+ |               |
|-------|----|--------|------|------|---------------|
| 8     | 43 | 2334.2 | 16.3 | 27.4 |               |
| 41    | 26 | 2038.7 | 17.1 | 30.0 | Volume <= 111 |
| 71    | 10 | 1871.7 | 18.1 | 32.0 | Volume <= 83  |
| ✓ 85  | 4  | 1831.7 | 19.0 | 32.5 | Model <= 73   |
| 62    | 6  | 1898.3 | 17.4 | 31.7 | Model > 73    |
| 22    | 16 | 2143.0 | 16.5 | 28.7 | Volume > 83   |
| 36    | 7  | 2098.1 | 18.0 | 28.6 | Model <= 73   |
| 12    | 9  | 2177.9 | 15.3 | 28.9 | Model > 73    |
| 26    | 4  | 2210.0 | 15.7 | 30.0 | Volume > 97   |
| 0     | 5  | 2152.2 | 15.0 | 28.0 | Volume <= 97  |
| -41   | 17 | 2786.1 | 15.1 | 23.5 | Volume > 111  |
| -9    | 5  | 2279.8 | 14.6 | 28.0 | Volume <= 116 |
| -54   | 12 | 2997.1 | 15.3 | 21.7 | Volume > 116  |
| -44   | 8  | 2921.1 | 15.8 | 22.5 | Model > 73    |
| -38   | 4  | 2563.0 | 14.0 | 25.0 | Clntrs <= 4   |
| -50   | 4  | 3279.2 | 17.4 | 20.0 | Clntrs > 4    |
| ✗ -75 | 4  | 3149.0 | 14.4 | 20.0 | Model <= 73   |



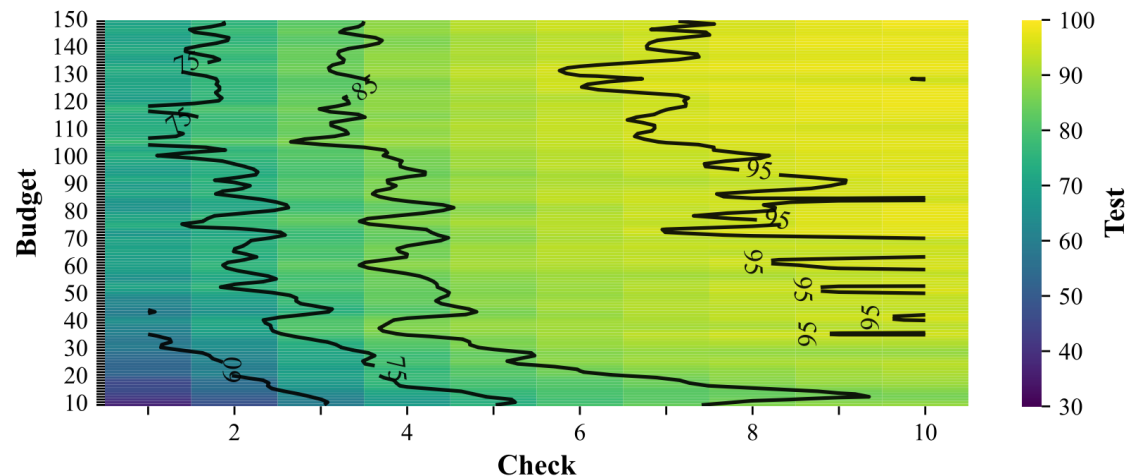


Q: how well does EZR generalize?

A: sort holdout via tree, check first few [31]

$\text{Win} = \text{int}(100 * (1 - \text{regret}))$

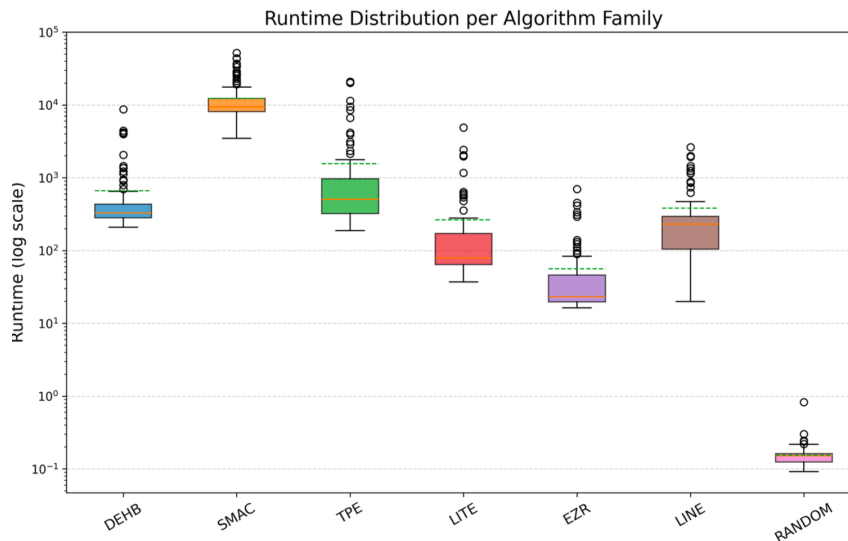
- 100,000 times
  - pick any MOOT data set
  - Train:
    - Active learning, labeling y rows with a **limited** budget.
    - Build tree from labels
  - Test:
    - Use tree to sort holdout
    - **Check** first few
- Active learning, very effective
  - Label 50 rows
  - Check 5
  - **Achieve** 85% of best



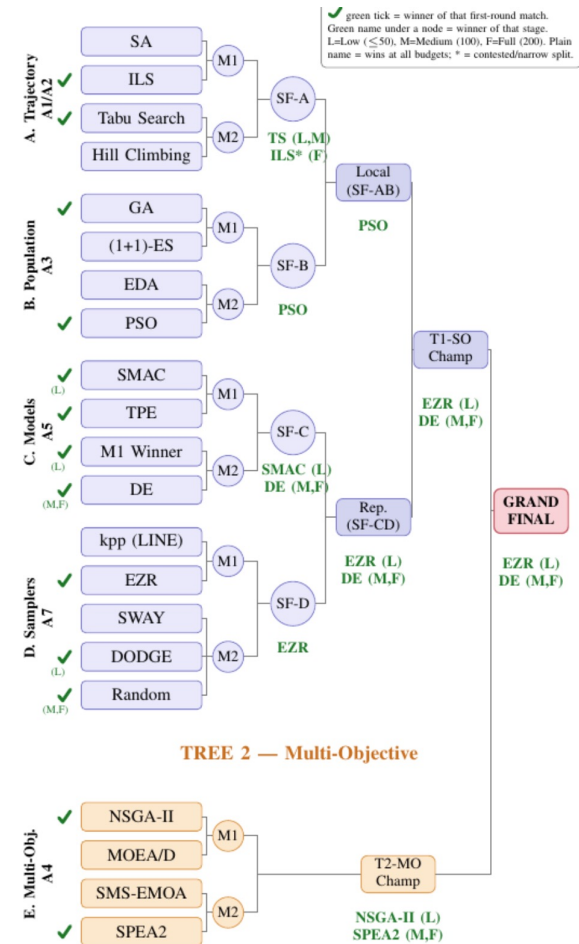
This entire 100,000-experiment study: one laptop, 2 hours, no cluster, no cloud bill.

Q: how fast and effective  
are EZR's trees?

A: very and very [32]



500x less CPU = 500x less energy. every run.

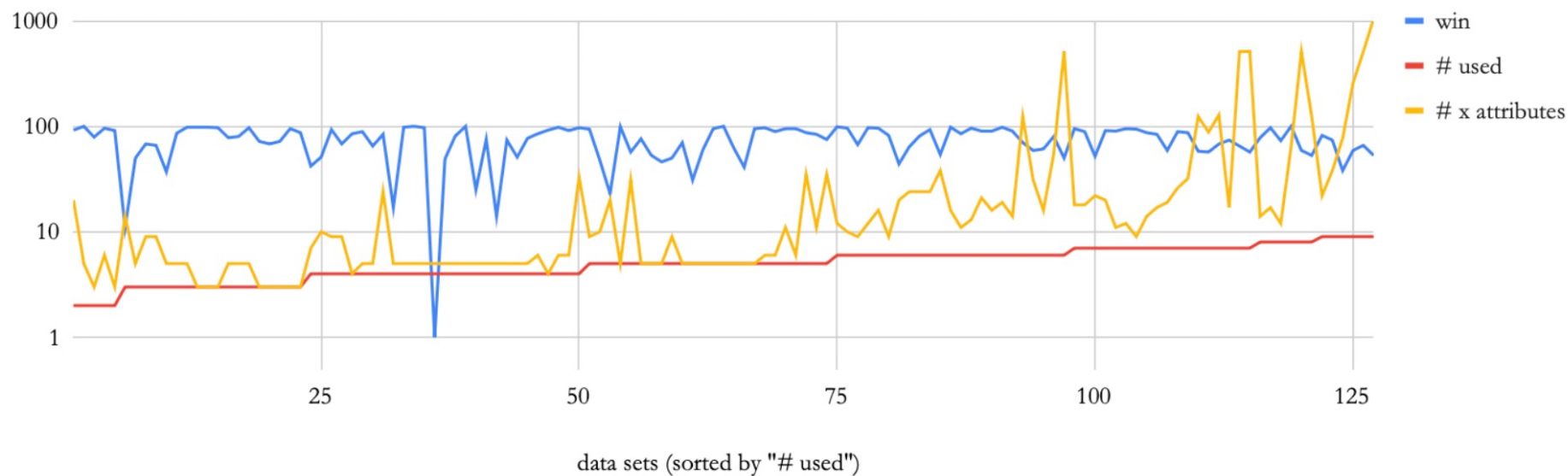
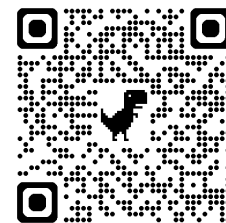


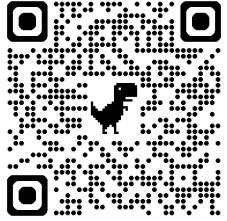
e.g. NSGA-II needs 1000 samples for what EZR reaches in 50

and are the models small enough to explain? →

Q: how big are EZR's models?

A: surprisingly small [33]





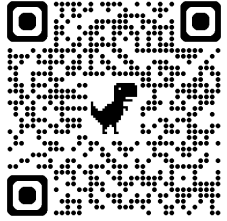
Q: so is EZR useful for explanation?

A: EZR's trees competitive with SOTA [33]

- Build trees using N **attributes**
- N= #attributes used in EZR's tree
- Using other tools: select **top** N attributes
- White = tied best
- Blue = within **top** 90%
- Yellow = within top 75%
- EZR goes a long way
  - E.g. last row
  - For a **3 goal** task
    - EZR uses 7 / 1044 attributes
    - EZR uses 150 rows
    - Arguably competitive
- **across 127 datasets: trees use**
- **4–9 of up to 1044 attributes**

| RLF | SHAP | EZR | anova | all | data        | x* | x    | y | budget | rows   |
|-----|------|-----|-------|-----|-------------|----|------|---|--------|--------|
| 95  | 100  | 100 | 100   | 100 | HSMGP       | 6  | 14   | 1 | 150    | 3457   |
| 87  | 81   | 82  | 82    | 85  | SS-R        | 7  | 14   | 2 | 150    | 3009   |
| 94  | 95   | 63  | 57    | 96  | SS-V        | 4  | 16   | 2 | 150    | 6841   |
| 74  | 74   | 59  | 74    | 99  | SS-W        | 7  | 16   | 2 | 150    | 65537  |
| 100 | 100  | 100 | 100   | 100 | X264-AM     | 6  | 16   | 1 | 150    | 1153   |
| 100 | 100  | 100 | 99    | 100 | SS-M        | 7  | 17   | 3 | 150    | 865    |
| 81  | 81   | 81  | 81    | 81  | SS-N        | 8  | 17   | 2 | 150    | 53663  |
| 59  | 62   | 53  | 52    | 58  | coc1000     | 9  | 20   | 5 | 150    | 1001   |
| 95  | 96   | 96  | 93    | 100 | SS-U        | 7  | 21   | 2 | 150    | 4609   |
| 98  | 98   | 93  | 98    | 98  | xomo_flight | 7  | 27   | 4 | 150    | 10001  |
| 92  | 92   | 90  | 90    | 92  | xomo_grnd   | 7  | 27   | 4 | 150    | 10001  |
| 94  | 95   | 95  | 95    | 95  | xomo_osp    | 7  | 27   | 4 | 150    | 10001  |
| 87  | 88   | 82  | 82    | 88  | xomo_osp2   | 8  | 27   | 4 | 150    | 10001  |
| 77  | 75   | 71  | 64    | 85  | SQL-AM      | 10 | 39   | 1 | 150    | 4654   |
| 87  | 91   | 92  | 74    | 94  | billing10k  | 7  | 88   | 3 | 150    | 10001  |
| 75  | 75   | 81  | 75    | 97  | Scrum100k   | 7  | 124  | 3 | 150    | 100001 |
| 82  | 88   | 92  | 88    | 97  | Scrum10k    | 8  | 124  | 3 | 150    | 10001  |
| 83  | 89   | 86  | 83    | 89  | Scrum1k     | 7  | 124  | 3 | 150    | 1001   |
| 81  | 99   | 93  | 81    | 99  | FFM-125     | 9  | 128  | 3 | 150    | 10001  |
| 81  | 97   | 89  | 97    | 97  | FFM-250     | 8  | 256  | 3 | 150    | 10001  |
| 95  | 93   | 93  | 95    | 95  | FFM-500     | 8  | 510  | 3 | 150    | 10001  |
| 81  | 96   | 87  | 96    | 96  | FM-500-1    | 8  | 511  | 3 | 150    | 10001  |
| 86  | 91   | 93  | 91    | 97  | FM-500-2    | 7  | 511  | 3 | 150    | 10001  |
| 95  | 95   | 95  | 87    | 95  | FM-500-3    | 9  | 513  | 3 | 150    | 10001  |
| 97  | 97   | 93  | 96    | 97  | FM-500-4    | 9  | 517  | 3 | 150    | 10001  |
| 82  | 94   | 94  | 71    | 100 | FFM-1000    | 7  | 1044 | 3 | 150    | 10001  |

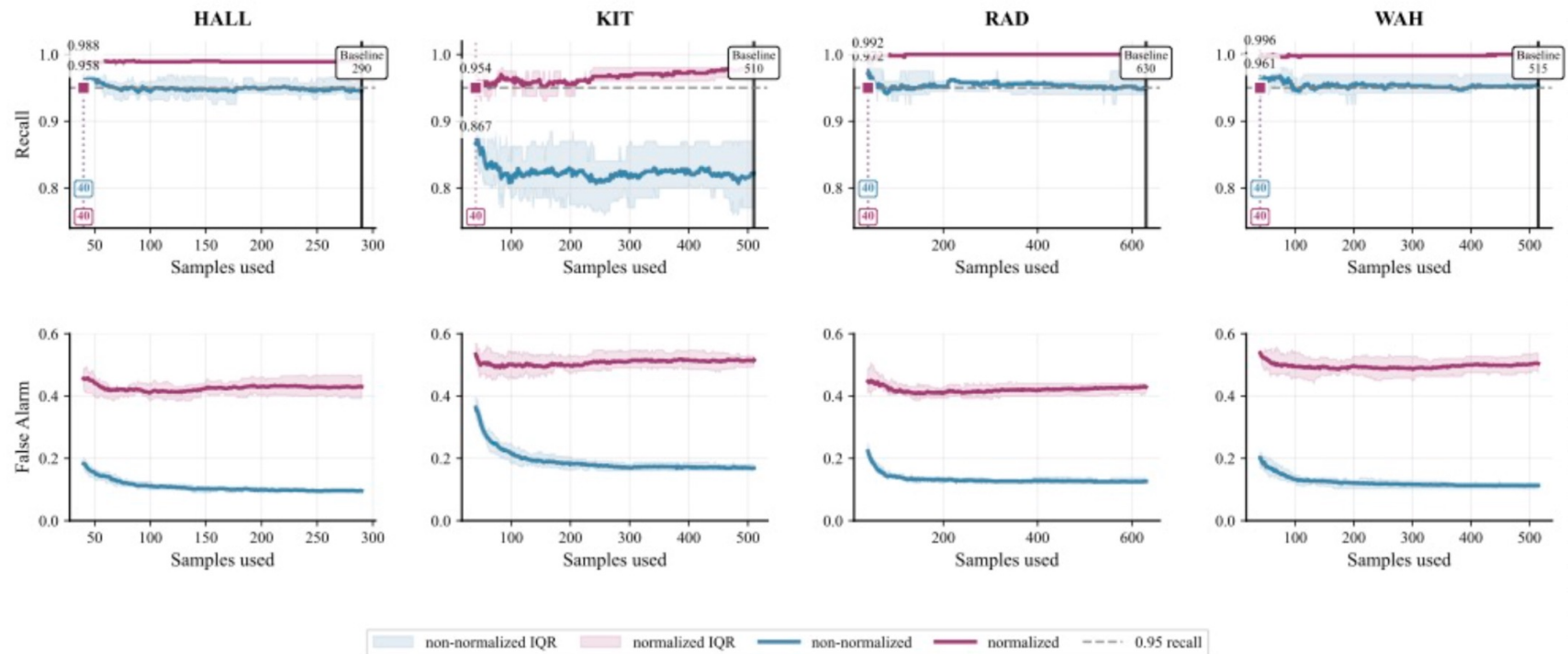
so why isn't everyone doing this? →



Q: what does EZR say about LLMs?

A: LLMs rarely tested against simpler tool [34]

- Search google scholar for **relevant** research



why easier AI?

how to do easier AI?

**why not more of easier AI?**

Q: if so good, why not more easier AI?

A: (weak) empirical standards [34] [35]



- In a survey of 229 SE papers on **LLMs in SE**
  - Only 5% compared to simpler approaches
- **Methodological** error
  - Other methods can produce results that are better and/or **faster**
    - (See above)
- My career:
  - Step 1: compare "it" vs a **stupid** straw man
  - Step 2: throw "it" away, use the straw man

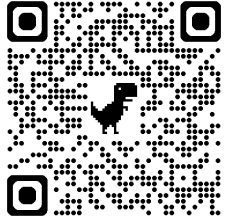


*also, the new toys are seductive →*



Q: if so good, why not more easier AI?

A: cause “big” sells; prompts/loops are cool [36]



- Welcome to my data center
  - ribbon on the door the mayor can cut
  - **such** a spectacle!
- Compare that to **timmm inc.**
  - No special software to sell
    - you can build it yourself
  - No special hardware to sell
    - our algorithms are so fast, they don't **need** it
  - IPO : low evaluation

The screenshot shows a terminal window with a dark background. The top part displays Python code for a data processing script. The bottom part shows a table of statistics. To the right of the terminal is a Claude Code interface with a red robot icon and text about MCP servers and usage limits.

```
[nothing] 0:htop*
15 if (s[1:] if s[1:] = "-" else s.isdigit(): return int(s)
16 try: return float(s)
17 except ValueError: return s="True" or (s!="False" and s)
18
19 def settings(s):
20     "Parse every var=val in a string into an o (vals coerced)."
21     pairs = re.findall(r"(\w+)=\S+", s)
22     return o(**{k: thing(v) for k, v in pairs})
23
24 def csv(file, clean=lambda s: s.partition("#")[0].split(","):
25     "Yield typed rows from a CSV file ('#' starts a comment)."
26     with open(file, encoding="utf-8") as f:
27         for line in f:
28             row = [x.strip() for x in clean(line)] # strip once, here
29             if any(row): # skip blank/comment lines
30                 yield tuple(thing(x) for x in row) # hashable: rows key caches
31
32 def say(x, dec=2):
33     "Pretty string: whole floats as ints, else `dec` places."
34     if isa(x, float):
35         return str(int(x)) if x == int(x) else f"{x:.{dec}f}"
36     if isa(x, o): x = vars(x) # unwrap a namespace
37     if isa(x, dict): return "{" + ", ".join(f"{k}: {say(v,dec)}"
38                                     for k, v in sorted(x.items())) + "}"
39
~/gits/aiez/nuff/nuff.py 24,30 3%
```

|      | 0           | 1           | 2           | 3           | 4           | 5           | 6           | 7           | 8           | 9           |
|------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Time | 24.5%       | 16.0%       | 7.9%        | 5.3%        | 2.0%        | 24.5%       | 16.0%       | 7.9%        | 5.3%        | 2.0%        |
| Mem  | 17.6G/24.0G | 17.6G/24.0G | 17.6G/24.0G | 17.6G/24.0G | 17.6G/24.0G | 17.6G/24.0G | 17.6G/24.0G | 17.6G/24.0G | 17.6G/24.0G | 17.6G/24.0G |
| Swp  | 1.40G/3.00G | 1.40G/3.00G | 1.40G/3.00G | 1.40G/3.00G | 1.40G/3.00G | 1.40G/3.00G | 1.40G/3.00G | 1.40G/3.00G | 1.40G/3.00G | 1.40G/3.00G |

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit

"timms-MacBook-Air.loc" 20:27 01-Jul-26

**Claude Code** v2.1.197  
Opus 4.8 · Claude Max  
~/gits/timm/misc/26/ssbse

▲ 3 MCP servers need authentication · run /mcp

Fable 5 is back.  
Until July 7, you can use up to 50% of your plan's weekly usage limit on Fable 5. If you hit your limit, you can continue on Fable 5 with usage credits. Fable 5 draws down usage faster than Opus 4.8. [Learn more](#)

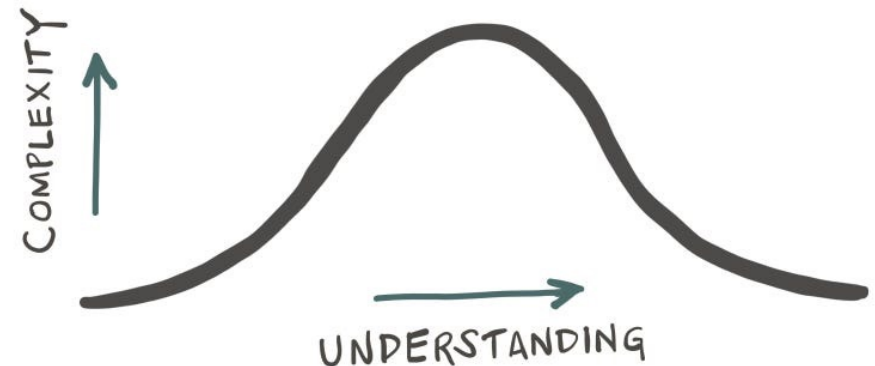
dir:~/gits/timm/misc/26/ssbse branch:maste...  
← for agents

# Q: if so good, why not easier AI?

## A: the simplicity paradox



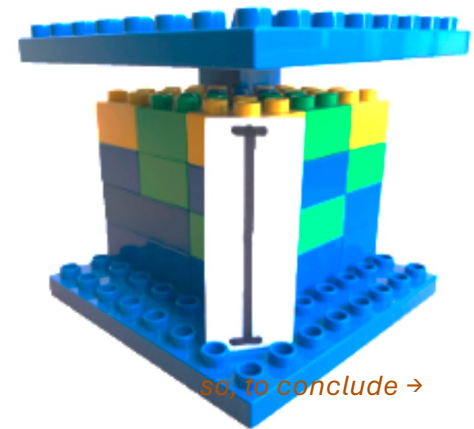
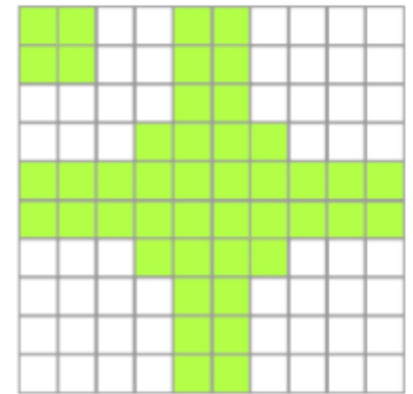
- Simplicity needs **experience**.
  - “Experience is the name every one gives to their **mistakes**.”  
— *Lady Windermere’s Fan* (1892)
- Simplicity is complicated to certify
  - You have to build and/or **run** simpler “it” and complex “it”
  - Needs lot of CPU



Q: if so good, why not more easier AI?  
A: human cognitive bias [37]



- Our simpler methods have not been **previously** explored
  - since other researchers **were not looking for them.**
- Why?
- Humans are biased to bloat
  - In **studies** of 100s of human subjects
    - Humans favor additive to subtractive changes
    - 1,155 to 297 (about 4:1)



**TL;DR**



# Lehman's **2nd** law <sup>[1]</sup>

- “as a system evolves, its complexity **increases** unless work is done to maintain or reduce it”
- Q: why / how can we **reduce** the complexity of AI?
- A: we must / we can
- **do**: baseline the simple:  
    `pip install ezr + tiny.cc/moot`
- **review**:
  - ask “compared to what simpler?”
- **teach**:
  - easier AI first; scale only when simple fails



/ (question | comment)\* /

code: *pip install ezr*  
data: *<http://tiny.cc/moot>*

## References

- [1] Lehman, Meir M. "Programs, Life Cycles, and Laws of Software Evolution." *Proceedings of the IEEE*, vol. 68, no. 9, 1980, pp. 1060-1076.
- [2] Nair V, **Menzies** T., Chen J. An (Accidental) Exploration of Alternatives to Evolutionary Algorithms for SBSE" presented at SSBSE 2016
- [3] Wirth, Niklaus. "A Plea for Lean Software." *Computer*, vol. 28, no. 2, 1995, pp. 64-68.
- [4] Bain & Company. *Global Technology Report 2025*. Bain & Company, Sept. 2025.
- [5] Schwartz, Roy, et al. "Green AI." *Communications of the ACM*, vol. 63, no. 12, 2020, pp. 54-63.
- [6] Xu, Tianyin, et al. "Hey, You Have Given Me Too Many Knobs! Understanding and Dealing with Over-Designed Configuration in System Software." *Proceedings of the 10th Joint Meeting on Foundations of Software Engineering (FSE)*, ACM, 2015, pp. 307-319.
- [7] Villalobos, Pablo, et al. "Will We Run Out of Data? Limits of LLM Scaling Based on Human-Generated Data." *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [8] Tawosi, Vali, Rebecca Moussa, and Federica Sarro. "Agile Effort Estimation: Have We Solved the Problem Yet? Insights from a Replication Study." *IEEE Transactions on Software Engineering*, vol. 49, no. 4, 2023, pp. 2677-2697.
- [9] Majumder, Suvodeep, Tim **Menzies** et al. "500+ Times Faster than Deep Learning: A Case Study Exploring Faster Methods for Text Mining StackOverflow." *Proceedings of the 15th International Conference on Mining Software Repositories (MSR)*, ACM, 2018.
- [10] Ling, Xiao, Tim **Menzies** et al. "Trading Off Scalability, Privacy, and Performance in Data Synthesis." *IEEE Access*, vol. 12, 2024, pp. 26642-26654.
- [11] Fu, Wei, and Tim **Menzies**. "Easy over Hard: A Case Study on Deep Learning." *Proceedings of the 11th Joint Meeting on Foundations of Software Engineering (ESEC/FSE)*, ACM, 2017, pp. 49-60.
- [12] Fu, Wei, Tim **Menzies**, and Xipeng Shen. "Tuning for Software Analytics: Is It Really Necessary?" *Information and Software Technology*, vol. 76, 2016, pp. 135-146.
- [13] Somvanshi, Shriyank, et al. "A Survey on Deep Tabular Learning." *arXiv preprint arXiv:2410.12034*, 2024.

## References (cont.)

- [14] Grinsztajn, Leo, Edouard Oyallon, and Gael Varoquaux. "Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?" Proceedings of NeurIPS Datasets and Benchmarks Track, 2022.
- [15] Pearson, Karl. "On Lines and Planes of Closest Fit to Systems of Points in Space." The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science, vol. 2, no. 11, 1901, pp. 559-572.
- [16] Amarel, Saul. "Program Synthesis as a Theory Formation Task: Problem Representations and Solution Methods." Machine Learning: An Artificial Intelligence Approach, Morgan Kaufmann, 1986.
- [17] Chang, Chin-Liang. "Finding Prototypes for Nearest Neighbor Classifiers." IEEE Transactions on Computers, vol. C-23, no. 11, 1974, pp. 1179-1184.
- [18] Johnson, William B., and Joram Lindenstrauss. "Extensions of Lipschitz Mappings into a Hilbert Space." Contemporary Mathematics, vol. 26, 1984, pp. 189-206.
- [19] De Kleer, Johan. "An Assumption-Based TMS." Artificial Intelligence, vol. 28, no. 2, 1986, pp. 127-162.
- [20] Crawford, James M., and Andrew B. Baker. "Experimental Results on the Application of Satisfiability Algorithms to Scheduling Problems." Proceedings of the 12th National Conference on Artificial Intelligence (AAAI), 1994, pp. 1092-1097.
- [21] Olshausen, Bruno A., and David J. Field. "Emergence of Simple-Cell Receptive Field Properties by Learning a Sparse Code for Natural Images." Nature, vol. 381, 1996, pp. 607-609.
- [22] Kohavi, Ron, and George H. John. "Wrappers for Feature Subset Selection." Artificial Intelligence, vol. 97, no. 1-2, 1997, pp. 273-324.
- [23] Williams, Ryan, Carla P. Gomes, and Bart Selman. "Backdoors to Typical Case Complexity." Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI), 2003, pp. 1173-1178.
- [24] Zhu, Xiaojin. "Semi-Supervised Learning Literature Survey." Computer Sciences Technical Report 1530, University of Wisconsin-Madison, 2005.
- [25] Settles, Burr. "Active Learning Literature Survey." Computer Sciences Technical Report 1648, University of Wisconsin-Madison, 2009.
- [26] **Menzies**, Tim. "Shockingly Simple: 'Keys' for Better AI for SE." IEEE Software, vol. 38, no. 2, 2021, pp. 114-118.



## References (cont.)

- [27] Nair, Vivek, Tim **Menzies**, et al. "Finding Faster Configurations Using FLASH." IEEE Transactions on Software Engineering, vol. 46, no. 7, 2018, pp. 794-811.
- [28] Yang, Chuanpeng, et al. "Survey on Knowledge Distillation for Large Language Models: Methods, Evaluation, and Application." ACM Transactions on Intelligent Systems and Technology, 2024.
- [29] Lindauer, Marius, et al. "SMAC3: A Versatile Bayesian Optimization Package for Hyperparameter Optimization." Journal of Machine Learning Research, vol. 23, no. 54, 2022, pp. 1-9.
- [30] **Menzies**, Tim, et al. "MOOT: A Repository of Many Multi-Objective Optimization Tasks." Proceedings of the International Conference on Mining Software Repositories (MSR), ACM, 2026.
- [31] Ganguly, Kishan Kumar, and Tim **Menzies**. "How Low Can You Go? The Data-Light SE Challenge." Proceedings of the ACM on Software Engineering, vol. 3, no. FSE, Article FSE185, 2026.
- [32] Ganguly, Kishan Kumar, and Tim **Menzies**. "Which Optimizer, At What Budget? A Tournament of Optimizers for Search-Based SE." 2026, under review.
- [33] Rayegan, Amirali, and Tim **Menzies**. "Minimal Data, Maximum Clarity: A Heuristic for Explaining Optimization." Journal of Systems and Software, vol. 238, 2026, 112897.
- [34] **Menzies**, Tim, and Srinath Srinivasan. "Can AI Be Easy? Lessons Learned from the EZR.py Toolkit." arXiv preprint arXiv:2606.03640, 2026.
- [35] Senthilkumar, Lohith, and Tim **Menzies**. "Can Large Language Models Improve SE Active Learning via Warm-Starts?" ACM Transactions on Software Engineering and Methodology, 2026.
- [36] Srinivasan, Srinath, and Tim **Menzies**. "Better Together, in the Right Order: Classical-then-LLM Optimization for SE." arXiv preprint, 2026.
- [37] Adams, Gabrielle S., et al. "People Systematically Overlook Subtractive Changes." Nature, vol. 592, 2021, pp. 258-261.